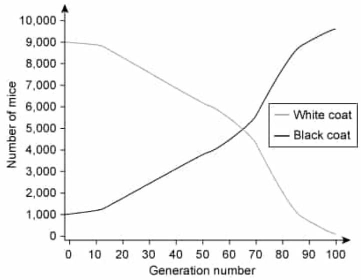
In an island population of mice, 10,000 mice are randomly sampled every 10 generations and their coat colors are recorded for a scientific study. The data for 100 generations of mice are shown in the graph below.

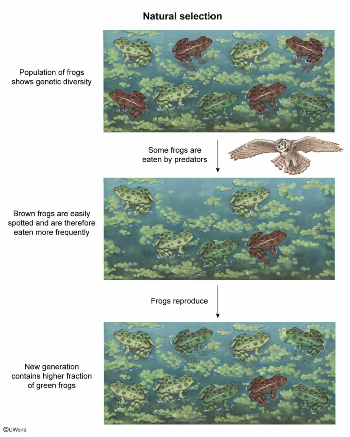


Which of the following conclusions about the black and white mice in this population is best supported by the graph?

- A. A population bottleneck caused the loss of white mice from the population but did not affect black mice. [5%]
- B. A mating event among white mice produced black mice that went on to have higher reproductive success than white mice. [10%]
- ✔ C. An environmental change resulted in natural selection for black mice but against white mice. [81%]
- D. A speciation event caused the lineage of white mice to split into a new species of black mice and decrease in frequency over time. [2%]

Correct
82% answered correctly

Explanation:



Natural selection is an evolutionary process in which beneficial traits that improve a species' fitness are more likely to be passed on to the next generation than less favorable traits. Accordingly, beneficial traits become more common and deleterious traits become less common with each generation, allowing the species to adapt to its environment gradually.

In this study, the coat colors of 10,000 mice were recorded every 10 generations. The graph shows that over 100 generations, the frequency of black mice *gradually* **increased** but the frequency of white mice **decreased**. Of all the answer choices, the conclusion that is best supported by the graph is that an environmental change resulted in natural selection that *improved* the survival and reproduction of black mice compared to white mice. Because such a change occurred gradually (**over many generations**), it is most likely that black coat color was selected for and that white coat color was selected against.

(Choice A) Bottleneck events are sudden environmental changes that rapidly (*not* gradually) decrease the number of individuals in a population, often leading to reduced genetic diversity. If white mice in this population were lost due to a bottleneck event, their frequency would decrease in a short period (*not* over 100 generations). In addition, bottleneck events affect *all* members of the population *randomly* (regardless of genotype or phenotype). Therefore, a bottleneck event *would also likely* affect the frequency of black mice in the population.

(Choices B and D) Because the graph shows that *both* white and black mice were already present at generation 0, these data *cannot* be used to determine if black mice *originated from* white mice. Therefore, there is *no* evidence to support that black mice were derived from a mating event among white mice or from a speciation event (ie, formation of a new species of black mice from an existing white species of mice).

Educational objective:
In natural selection, beneficial traits that improve fitness are more likely to be passed to subsequent generations than less favorable traits. Beneficial traits should become more common with each generation, allowing the species to adapt to its environment.

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